WORKED SOLUTION

STEP II 2024, Question 2

Pure • Series and Integration

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Part (i) turns every integral $\int_0^1 x^m/(8+x^3)\,\mathrm{d}x$ into a series, so any series built from the same ingredients is a combination of these integrals. To evaluate a series in (ii) or (iii), split its bracket into terms of the form $1/(3k+m+1)$, recombine the matching integrands into one fraction, and integrate. The factorisation $8+x^3 = (x+2)(x^2-2x+4)$ is what makes the resulting integrals tractable.

Solution

Throughout, write

$$I_m = \int_0^1 \frac{x^m}{8+x^3}\,\mathrm{d}x.$$

Part (i)

Taking out the factor 8 and using $(1+u)^{-1} = \sum_{k\geq 0}(-1)^k u^k$ for $|u|<1$ with $u = x^3/8$,

$$(8+x^3)^{-1} = \frac{1}{8}\left(1+\frac{x^3}{8}\right)^{-1} = \sum_{k=0}^{\infty}\frac{(-1)^k x^{3k}}{8^{k+1}} = \sum_{k=0}^{\infty}\frac{(-1)^k}{2^{3(k+1)}}\,x^{3k}.$$

This is valid when $|x^3/8|<1$, that is, when $|x|<2$.

Now multiply by x^m and integrate term by term over $[0,1]$, which lies inside the interval of validity (and we are told not to worry about convergence):

$$I_m = \sum_{k=0}^{\infty}\frac{(-1)^k}{2^{3(k+1)}}\int_0^1 x^{3k+m}\,\mathrm{d}x = \sum_{k=0}^{\infty}\frac{(-1)^k}{2^{3(k+1)}}\cdot\frac{1}{3k+m+1},$$

since $\int_0^1 x^n\,\mathrm{d}x = 1/(n+1)$ with $n = 3k+m \geq 0$. That is the required result.

Part (ii)

The three denominators $3k+3$, $3k+2$ and $3k+1$ are $3k+m+1$ with $m = 2, 1, 0$ respectively. So by part (i) the series is

$$I_2 - 2I_1 + 4I_0 = \int_0^1 \frac{x^2 - 2x + 4}{8 + x^3}\,\mathrm{d}x.$$

? **Ask yourself** Why should $x^2 - 2x + 4$ cancel with anything?

Because $8 + x^3$ is a sum of two cubes: $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$ with $a = x$, $b = 2$ gives

$$8 + x^3 = (x+2)(x^2 - 2x + 4).$$

Hence the integrand simplifies to $\dfrac{1}{x+2}$, so the series equals $\displaystyle\int_0^1 \frac{1}{x+2}\,\mathrm{d}x$, as required. Finally,

$$\int_0^1 \frac{1}{x+2}\,\mathrm{d}x = \Big[\ln(x+2)\Big]_0^1 = \ln 3 - \ln 2 = \ln \tfrac{3}{2}.$$

Part (iii)

First split the bracket into partial fractions in k. Write $2k+1 = \alpha(3k+2) + \beta(3k+1)$. Putting $k = -\frac{1}{3}$ gives $\frac{1}{3} = \alpha$; putting $k = -\frac{2}{3}$ gives $-\frac{1}{3} = -\beta$, so $\beta = \frac{1}{3}$. Therefore

$$\frac{72(2k+1)}{(3k+1)(3k+2)} = \frac{24}{3k+1} + \frac{24}{3k+2},$$

and by part (i) (with $m = 0$ and $m = 1$) the series equals

$$24I_0 + 24I_1 = \int_0^1 \frac{24(1+x)}{(x+2)(x^2 - 2x + 4)}\,\mathrm{d}x.$$

Now split this integrand into partial fractions:

$$\frac{1+x}{(x+2)(x^2 - 2x + 4)} = \frac{A}{x+2} + \frac{Bx + C}{x^2 - 2x + 4},$$

so that

$$1 + x = A(x^2 - 2x + 4) + (Bx + C)(x+2).$$

Putting $x = -2$ gives $-1 = 12A$, so $A = -\frac{1}{12}$. Comparing coefficients of x^2 gives $0 = A + B$, so $B = \frac{1}{12}$. Comparing constant terms gives $1 = 4A + 2C$, so $C = \frac{2}{3}$. (The x coefficients check: $-2A + 2B + C = \frac{1}{6} + \frac{1}{6} + \frac{2}{3} = 1$.) Multiplying by 24,

$$\begin{aligned}\frac{24(1+x)}{8+x^3} &= -\frac{2}{x+2} + \frac{2x+16}{x^2 - 2x + 4} \\ &= -\frac{2}{x+2} + \frac{2x-2}{x^2 - 2x + 4} + \frac{18}{(x-1)^2 + 3},\end{aligned}$$

where we have split off the derivative $2x-2$ of the quadratic and completed the square in what remains.

We integrate each piece over $[0,1]$.

$$\int_0^1 \frac{-2}{x+2}\,\mathrm{d}x = -2\ln\tfrac{3}{2} = -2\ln 3 + 2\ln 2,$$
$$\int_0^1 \frac{2x-2}{x^2-2x+4}\,\mathrm{d}x = \Big[\ln(x^2-2x+4)\Big]_0^1 = \ln\tfrac{3}{4} = \ln 3 - 2\ln 2,$$
$$\int_0^1 \frac{18}{(x-1)^2+3}\,\mathrm{d}x = \frac{18}{\sqrt{3}}\Big[\arctan\frac{x-1}{\sqrt{3}}\Big]_0^1$$
$$= 6\sqrt{3}\Big(0 - \arctan(-\tfrac{1}{\sqrt{3}})\Big) = 6\sqrt{3}\cdot\frac{\pi}{6} = \pi\sqrt{3}.$$

In the last line we used $\int \frac{\mathrm{d}u}{u^2+c^2} = \frac{1}{c}\arctan\frac{u}{c}$ with $u = x-1$, $c = \sqrt{3}$. Adding the three pieces, the $\ln 2$ terms cancel and

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2^{3(k+1)}}\left(\frac{72(2k+1)}{(3k+1)(3k+2)}\right) = \pi\sqrt{3} - \ln 3,$$

so $a = 3$ and $b = 3$.

Checks

- The series converge very fast (each term is at most an eighth of the previous one), so a few terms are a good check. In (ii), the first three terms of the series give $0.41667 - 0.01198 + 0.00084 \approx 0.40553$, against $\ln\frac{3}{2} \approx 0.40547$.
- In (iii), $\pi\sqrt{3} - \ln 3 \approx 5.4414 - 1.0986 = 4.3428$; summing the first 200 terms of the series by computer agrees to twelve decimal places. Even the $k = 0$ term alone gives $\frac{1}{8}\cdot 36 = 4.5$, the right size.

Common slips

- Writing the expansion with 8^{-k} instead of $8^{-(k+1)}$: the factor $\frac{1}{8}$ taken out at the start must not be lost.
- Matching $\frac{1}{3k+3}$ with I_3 rather than I_2: the denominator is $3k+m+1$, so m is one less than the constant.
- Forgetting that $\arctan\big(-\frac{1}{\sqrt{3}}\big)$ is negative, which gives the wrong sign for the $\pi\sqrt{3}$ term.

Beyond STEP

Swapping a sum and an integral, as in part (i), is legitimate here because the power series converges uniformly on $[0,1]$; when such swaps are allowed is a central theme

of first-year analysis. The same method sums $1 - \frac{1}{4} + \frac{1}{7} - \frac{1}{10} + \cdots = \int_0^1 \frac{\mathrm{d}x}{1+x^3}$ exactly as in part (iii). Abbott's *Understanding Analysis* explains uniform convergence and term-by-term integration carefully.

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